

Evaluating Limits by Factoring and Rationalizing

Answer Key

AP Calculus AB/BC

Quick Answers

1. 8
2. 1
3. $\frac{1}{6}$
4. $\frac{5}{6}$

5. $\frac{1}{4}$
 6. 12
 7. (a) the limit = 3, —
 8. 4
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What is verified

8 of 9 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problem 7. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

LIM-1 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 9 tagged checks.

Common wrong answers

4. If they got 1: cancelled the x squared terms across the subtraction, leaving $(-x-6)/(-9)$, then substituted $x = 3$

6. If they got 0: expanded the cube as 8 plus h cubed, so the quotient collapsed to h squared

1. Table of $f(x) = \frac{x^2 - 16}{x - 4}$ at $x = 3.9, 3.99, 4.01, 4.1$ gives 7.9, 7.99, 8.01, 8.1. Predict $\lim_{x \rightarrow 4} f(x)$, then confirm by factoring and cancelling.

Solution: The table values close in on 8 from both sides, so the prediction is 8. Direct substitution gives $\frac{0}{0}$, so factor:

$$\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4 \quad (x \neq 4).$$

The cancelled form $x + 4$ agrees with f everywhere except at $x = 4$, where f has a hole. Limits ignore the single missing point, so substitute into $x + 4$: $\lim_{x \rightarrow 4} f(x) = 4 + 4$. 8

2. Evaluate $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.

Solution: Substituting $x = -2$ gives $\frac{4-10+6}{0} = \frac{0}{0}$, so the numerator must contain the factor $x + 2$:

$$\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3 \quad (x \neq -2).$$

Now substitute: $-2 + 3$.

1

3. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$.

Solution: Substitution gives $\frac{3-3}{0} = \frac{0}{0}$. There is nothing to factor, so multiply numerator and denominator by the conjugate $\sqrt{x+9} + 3$:

$$\begin{aligned} \frac{\sqrt{x+9} - 3}{x} \cdot \frac{\sqrt{x+9} + 3}{\sqrt{x+9} + 3} &= \frac{(x+9) - 9}{x(\sqrt{x+9} + 3)} \\ &= \frac{x}{x(\sqrt{x+9} + 3)} = \frac{1}{\sqrt{x+9} + 3} \quad (x \neq 0). \end{aligned}$$

Substituting $x = 0$ gives $\frac{1}{3+3}$.

$\frac{1}{6}$

4. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$.

Solution: Substitution gives $\frac{0}{0}$, so factor both parts completely — the cancelling factor is $x - 3$ in each:

$$\frac{x^2 - x - 6}{x^2 - 9} = \frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \quad (x \neq 3).$$

Substituting $x = 3$ gives $\frac{5}{6}$.

$\frac{5}{6}$

5. Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$.

Solution: Substitution gives $\frac{0}{0}$. Multiply by the conjugate of the numerator, $\sqrt{x} + 2$:

$$\frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} = \frac{x - 4}{(x - 4)(\sqrt{x} + 2)} = \frac{1}{\sqrt{x} + 2} \quad (x \neq 4).$$

(The same cancellation follows from factoring $x - 4 = (\sqrt{x} - 2)(\sqrt{x} + 2)$.) Substituting $x = 4$ gives $\frac{1}{2+2}$.

$\frac{1}{4}$

6. Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^3 - 8}{h}$, the instantaneous rate of change of $y = x^3$ at $x = 2$.

Solution: Expand the cube in full — this is the step the common error skips:

$$(2 + h)^3 = 8 + 12h + 6h^2 + h^3.$$

So the numerator is $12h + 6h^2 + h^3 = h(12 + 6h + h^2)$ and

$$\frac{(2 + h)^3 - 8}{h} = 12 + 6h + h^2 \quad (h \neq 0).$$

Letting $h \rightarrow 0$ leaves the constant term.

12

7. Table of $g(x) = \frac{x^3 - 8}{x^2 - 4}$ at $x = 1.9, 1.99, 2.01, 2.1$ gives 2.9256, 2.9925, 3.0075, 3.0756.

(a) Give the exact value of $\lim_{x \rightarrow 2} g(x)$. **(b)** Explain why the table entries alone do not prove that value.

Solution (a): Substitution gives $\frac{0}{0}$. Use the difference-of-cubes and difference-of-squares factorings:

$$\frac{x^3 - 8}{x^2 - 4} = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)} = \frac{x^2 + 2x + 4}{x + 2} \quad (x \neq 2).$$

Substituting $x = 2$ gives $\frac{4+4+4}{4} = \frac{12}{4}$.

3

Solution (b) — instructor-judged, no machine check. A correct response must make the sampling point: the table reports four inputs chosen by the author, and no finite list of nearby inputs can rule out different behaviour strictly between them or exactly at $x = 2$, so the column is evidence that the values are heading toward 3 and not a proof that they do. Full credit also names what does prove it — cancelling the common factor $x - 2$ turns g into a function that is continuous at 2, so its limit is its value there. Accept a response that makes the finitely-many-samples point even if the second half is only implied; do not accept “because the numbers round to 3.”

8. Challenge. Evaluate $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x + 3} - 2}$.

Solution: Substitution gives $\frac{0}{0}$. The radical sits in the denominator, so multiply above and below by the denominator’s conjugate, $\sqrt{x + 3} + 2$:

$$\begin{aligned} \frac{x - 1}{\sqrt{x + 3} - 2} \cdot \frac{\sqrt{x + 3} + 2}{\sqrt{x + 3} + 2} &= \frac{(x - 1)(\sqrt{x + 3} + 2)}{(x + 3) - 4} \\ &= \frac{(x - 1)(\sqrt{x + 3} + 2)}{x - 1} = \sqrt{x + 3} + 2 \quad (x \neq 1). \end{aligned}$$

Substituting $x = 1$ gives $\sqrt{4} + 2 = 2 + 2$.

4